

Force Feedback Button that Focuses on Both Operation and Non-Operation

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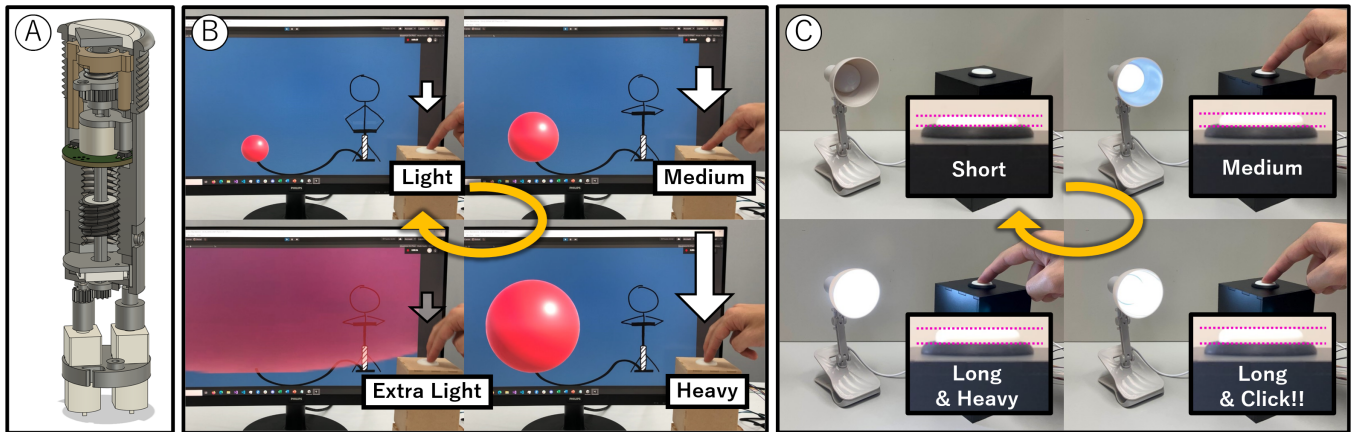


Figure 1: Proposed device and its applications. (a) Push button device with multiple force-feedback. (b) Application example during operation. User can inflate the balloon by pressing the button repeatedly. The button becomes heavier as the balloon inflates. (c) Application example during non-operation. The height of the button is changed with the brightness of the light.

Abstract

Push buttons are common input devices used in various places in everyday life, and various research projects have investigated interactions involving them. However, most existing research has been limited to interactions that occur during the operation of push buttons. In this paper, we propose a force feedback button. The button enables interactions during both operation and non-operation by physically driving push buttons, thereby providing a new modality for push buttons. The device can provide three types of force feedback: “pressing force,” “click feeling,” and “stroke.” Using this feedback, we developed application examples for operation and non-operation.

CCS Concepts

• Human-centered computing → Haptic devices.

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1 Introduction

Push buttons are common input devices used in various places in everyday life, and various research projects have investigated interactions involving them. For example, studies have presented various click sensations by providing vibration while the button is pressed [Ogawa et al. 2015][Park et al. 2020], as well as a push button simulator using a moving coil motor [Doerrer and Werthschützky 2002] or linear force actuator [Liao et al. 2020].

However, most existing research has been limited to interactions during button operation. Therefore, in this paper, we propose a force feedback button that enables interactions during both operation and non-operation by physically driving push buttons (Fig. 1). Although there are methods to indicate button states using OELD¹, our methods is unique in that it drives the button itself.

2 Proposal

In this paper, we propose a force feedback button that enables interactions during both operation and non-operation by physically driving push buttons, thereby providing a new modality for push buttons. During operation, the system can change the weight of the button and generate a click sensation at a certain height. During non-operation, the system can move the button up/down to draw attention, flattening the button to prevent operation.

We focused on three elements related to the sensation of pushing the button: pressing force, click feeling, and stroke. These elements

¹<https://www.artlebedev.com/optimus/maximus/>

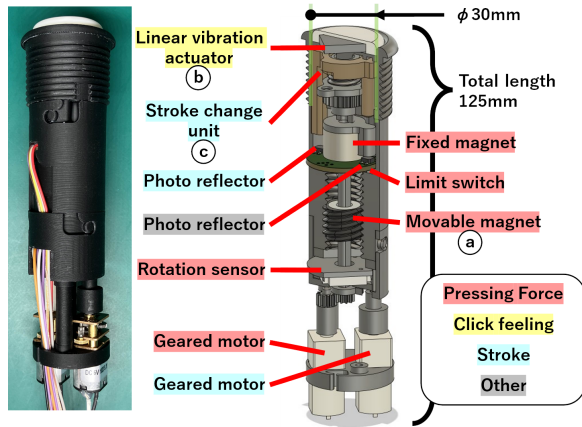


Figure 2: Left: Appearance of a push button-type device. Right: Internal structure of a push button-type device. The main applications for each type are color-coded.

were summarized by the authors based on the parameters of mechanical keyboards².

3 Implementation

The pressing force, click feeling, and stroke of a typical button are determined by the button design and mechanism, making it difficult to change them flexibly. In this section, we describe how these elements were expressed and implemented.

The appearance and internal structure of the push button-type devices are shown in Fig. 2. The casing and mechanical parts were fabricated using a 3D printer.

The pressing force was adjusted using the repulsive force between two permanent magnets. Two neodymium magnets were arranged coaxially, with their magnetic poles facing each other, and the pressing force was adjusted by moving one of the magnets up or down (Fig. 2 (a)). The position of the magnet was controlled by feedback from a geared motor and rotation sensor. The pressing force can be changed in 16 steps between approximately 0.1 and 3.7 N.

The click feeling was adjusted using a linear resonant actuator (LD14-002, as shown in Fig. 2 (b)). The actuator was fitted to the back of a button touched by a finger. Pressing was detected using a photoreflector. The click feeling can be selected from five types created in advance, and the height at which the vibration is generated can be changed in 16 steps.

The stroke was adjusted by physically limiting the button movement using a stroke change unit (Fig. 2 (c)). The position of the stroke change unit was controlled using a geared motor and photoreflector. The stroke can be controlled in eight steps between 3 and 0 mm (flat).

These three elements were controlled using a microcontroller (PIC18F27Q43). By incorporating external communication functionality (UART), it is possible to acquire the press status and modify each element from other microcontrollers or PCs.

4 Application Examples

We developed application examples during operation and non-operation.

The first application example is a game inspired by inflating a balloon (Fig. 1(b)), which focuses on button operations. This game was created using Unity. Pressing the button gradually inflates a balloon on the PC screen. Initially, the force required to press the button is small, but as the balloon inflates, the required force gradually increases. More force is required to further inflate the balloon. Once the balloon reaches a certain size, it bursts, and the force required to press the button returns to its minimum state, mimicking the sensation of air escaping.

The second application example is a lighting switch (Fig. 1(c)), which focuses on the button state during non-operation. The LED light is controlled by PWM from a PIC microcontroller. There are multiple brightness levels, which are changed by pressing the button (off → 1 → 2 → 3 → off). In the initial state (off), the stroke is short. As the brightness level increases, the stroke to raise the button lengthens. Users can estimate the status of the light based on the appearance of the button. In addition, when the brightness reaches its maximum level, a click is generated and the pressing force is increased. This informs the user that the brightness is at its maximum level.

5 Future Work

In the future, we plan to expand the system to control multiple buttons. We will then implement various applications for entertainment purposes, such as games and musical instruments, as well as for use in everyday life.

Acknowledgments

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²<https://www.cherry.de/en-gb/product/mx2a-red>